

PRODUCT REQUIREMENTS DOCUMENT

Development Works Monitoring Dashboard

Centralized Development & Construction Works Monitoring System

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1. Project Overview

नगर निगम मथुरा-वृंदावन द्वारा संचालित सभी Development एवं Construction Works की जानकारी को एक centralized system में consolidate करके उनकी वर्तमान स्थिति, financial source, ward, responsible JE, milestones, progress percentage और photographs को व्यवस्थित तरीके से monitor करने के लिए Development Works Monitoring Dashboard विकसित किया जाना है।

System का primary objective existing information को एक single monitoring layer में लाना है, ताकि Administrator, Municipal Officers, Junior Engineers (JE), Councillors तथा आवश्यकता अनुसार citizens को role-based और आसान तरीके से work status उपलब्ध हो सके।

Key Business Outcomes

- सभी development works के लिए centralized master record.
- Ward-wise, Department-wise और Scheme-wise monitoring.
- JE द्वारा field-level progress updates और photographic evidence.
- Milestone/Stage-wise execution tracking.
- Delayed और pending works की timely identification.
- Management-level KPI dashboard और downloadable reports.
- Future GIS, public dashboard और notification integrations के लिए scalable foundation.

2. Existing System & Data Sources

Source	Current Coverage	Role in New System
nnmv.online	SFC एवं नगर निगम निधि / Own Revenue से संबंधित works	Data source / integration or migration source
nnmvprojects.online	Central एवं State Government योजनाओं से संबंधित projects	Data source / integration or migration source
Construction Department Portal	Existing development/construction work information	Reference for existing fields, workflow and business logic

नए Dashboard को existing portals के आधार पर अधिक structured, user-friendly और progress-oriented monitoring system के रूप में विकसित किया जाएगा।

3. Objectives

- नगर निगम के सभी विकास कार्यों की single-window visibility.
- हर work की current status और progress percentage को track करना.
- JE द्वारा नियमित field updates capture करना.
- Before, Progress और Completion photographs को work history के साथ सुरक्षित रखना.

- Milestone-based execution tracking.
- Councillors को अपने ward के works की visibility देना.
- Officers को Nagar Nigam-wide consolidated monitoring देना.
- Search, filter और reporting के माध्यम से faster decision-making.
- Data integrity, accountability और auditability बढ़ाना.

4. Scope

In Scope

- Web-based centralized monitoring dashboard.
- Role-based login for Admin, Officers, JE और Councillors.
- Development Work Master.
- Scheme, Financial Source, Ward, Department, Work Type, Contractor और JE master data.
- Milestone management.
- Progress updates.
- Photo/document management.
- Dashboards and analytics.
- Search/filter.
- Reports and exports.
- Audit trail.
- API/data integration with existing portals, subject to availability.
- Future-ready GIS/public dashboard architecture.

Out of Scope for Initial MVP

- Full e-tendering/procurement workflow.
- Complete contractor billing/payment processing.
- Advanced AI-based photo verification.
- Native Android/iOS application; a responsive web/PWA approach can be considered first.

5. Financial Sources / Schemes

Category	Examples
Central Finance	15th Finance Commission
State Finance	State Finance Commission (SFC)
Nagar Nigam Fund	Own Revenue / Nagar Nigam Nidhi
Central Schemes	Swachh Bharat Mission, AMRUT आदि
State Schemes	State Government Schemes
Special Projects	CM-GRIDS आदि
Infrastructure	Sewerage, Drainage, Roads आदि
Other	Future / custom schemes

Scheme Master और Financial Source Master configurable होने चाहिए ताकि future में नई schemes को software code change के बिना add किया जा सके।

6. User Roles & Access Control

Role	Primary Access
Super Admin	Full system configuration, users, masters, works, reports, audit logs
Municipal Officer	Nagar Nigam-wide monitoring, analytics, reports, verification
Department Officer	Department-specific monitoring and reporting
JE / Junior Engineer	Assigned works: progress, milestones, photos, remarks, status updates
Councillor	Read-only monitoring of assigned ward works
Public/Citizen	Only approved public work information, if enabled

Role-Based Access Control Rules

- हर user को role और required scope/ward/department के साथ map किया जाएगा.
- JE केवल assigned works पर progress update कर सके.
- Councillor अपने assigned ward के works देख सके; default रूप से editing अधिकार न हों.
- Officers अपने authorized department/region या पूरे Nagar Nigam के data तक पहुँच सकें.
- Super Admin को full administrative control हो.

7. Development Work Master

प्रत्येक development work के लिए unique Work ID generate किया जाएगा।

Information Group	Fields
Basic	Work ID, Work Name, Description, Ward, Zone, Area/Location, Work Type
Scheme	Scheme, Financial Source, Work Category
Administrative	Approval No./Date, Technical Sanction No./Date, Work Order No./Date
Responsibility	Department, Assigned JE, Officer, Contractor/Agency
Financial	Estimated Cost, Sanctioned Amount, Work Order

	Amount, Expenditure, Balance
Timeline	Start Date, Original Completion Date, Revised Completion Date, Actual Completion Date
Location	Address/landmark, Latitude, Longitude — if GIS enabled

8. Milestone / Stage Management

हर work के लिए predefined या custom milestones configure किए जा सकेंगे।

Example — Road Work:

31. Base Preparation
32. Gitti / Sand Work
33. Interlocking / Surface Work
34. Finishing
35. Work Completed

हर milestone में: Milestone Name, Sequence, Target Date, Actual Start/Completion Date, Progress %, Status, Remarks और Photographs store किए जा सकेंगे।

Future enhancement: milestone weights के आधार पर overall work progress का automatic calculation।

9. Progress Update Module

JE field inspection या work execution के दौरान progress update create कर सकेगा।

Field	Requirement
Work ID	Mandatory
Update Date	Mandatory
Current Progress %	Mandatory
Milestone	Mandatory/Configurable
Status	Mandatory
Remarks	Mandatory
Photograph	At least one for field progress update, configurable
GPS	Optional / future-ready
Video	Optional / future-ready
Delay Reason	Required when status is Delayed, recommended
Next Target Date	Optional

10. Photo & Document Management

Photograph Categories

Category	Purpose
Before	कार्य शुरू होने से पहले की स्थिति
Progress	कार्य के दौरान की स्थिति
Completion	कार्य पूर्ण होने के बाद की स्थिति

Photo Metadata

- Work ID
- Photo category / milestone
- Upload date & time
- Uploaded by
- Remarks
- GPS location, if enabled
- File type and size validation

Photos को secure storage में रखा जाए और unauthorized replacement/deletion से बचाने के लिए audit trail maintain किया जाए।

11. Status Management

Status	Meaning
Not Started	कार्य प्रारंभ नहीं हुआ
In Progress	कार्य प्रगति पर है
On Hold	कार्य अस्थायी रूप से रुका है
Delayed	निर्धारित timeline से पीछे

Completed	कार्य पूर्ण
Cancelled	कार्य निरस्त
Under Verification	Completion verification pending

Status list configurable रखी जानी चाहिए।

12. Dashboards

12.1 Central Admin / Officer Dashboard

- Total Development Works
- Total Work Value
- Not Started
- In Progress
- Completed
- Delayed
- On Hold
- Average/Overall Completion %

12.2 Ward-wise Analytics

प्रत्येक Ward में total, ongoing, completed, delayed works और total sanctioned value का consolidated view।

12.3 Scheme-wise Analytics

15th Finance Commission, SFC, Nagar Nigam Fund, AMRUT, SBM, CM-GRIDS आदि के अनुसार work distribution और progress।

12.4 Department-wise Analytics

Construction/Engineering, Water, Sewerage, Drainage और अन्य configured departments के अनुसार monitoring।

13. Search & Filters

- Ward
- Zone
- Work ID
- Work Name
- Work Type
- Scheme
- Financial Source
- Department
- JE
- Contractor
- Status
- Progress %
- Start Date
- Completion Date

Example filter: Ward = 32 + Scheme = SFC + Status = In Progress → केवल matching works display हों।

14. Work Details Page

हर work के लिए dedicated detail page उपलब्ध होगा। Suggested information architecture:

WORK ID → Work Name → Ward → Scheme → Status → Overall Progress → Basic Information → Financial Information → Timeline → Milestones → Progress Updates → Before Photos → Progress Photos → Completion Photos → Remarks/Delay Reason → Assigned JE → Contractor → Documents

Work Details – Minimum Information

- Work ID and name
- Ward / zone / location
- Scheme and financial source
- Work type
- Sanctioned/work order amount
- JE and contractor
- Start/completion dates
- Current status and progress
- Milestone status
- Photographic history
- Relevant documents

15. GIS / Map Integration

System architecture में GIS integration के लिए location fields रखे जाएँ। प्रत्येक work को map पर status/progress के आधार पर visualize किया जा सके।

Map Indicator	Suggested Meaning
Green	Completed
Yellow	In Progress
Red	Delayed
Grey	Not Started

GIS को MVP में optional रखा जा सकता है, लेकिन database/API architecture में latitude-longitude fields शुरू से रखना recommended है।

16. Citizen / Public Dashboard

यदि Nagar Nigam public transparency enable करना चाहे तो approved work information public dashboard पर उपलब्ध कराई जा सकती है।

- Ward
- Work Name
- Location
- Scheme
- Work Cost
- Status
- Progress %
- Before/Progress/Completion Photos

Internal remarks, user information और sensitive administrative data public नहीं किया जाना चाहिए।

17. Reports & Exports

Report	Purpose
Ward-wise Work Report	Ward-level works and status
Scheme-wise Work Report	Scheme performance
Department-wise Report	Department monitoring
JE-wise Report	Assigned work and update performance
Delayed Works Report	Management escalation
Completed Works Report	Completion tracking
Financial Source Report	Funding-wise distribution
Monthly Progress Report	Periodic review
Work-wise Progress Report	Detailed work history
Photo Progress Report	Visual evidence

Export formats: Excel, CSV और PDF.

18. Notifications

- Work completion deadline approaching
- Work delayed
- JE progress update pending
- Work marked Completed
- Completion verification pending
- New work assigned to JE

Initial implementation में in-app notifications recommended हैं। Email/SMS/WhatsApp integration को future phase में रखा जा सकता है।

19. Audit Trail

Government monitoring system के लिए महत्वपूर्ण actions का immutable/traceable audit history maintain किया जाना चाहिए।

Audit Item	Example
User	JE001
Action	Progress Updated
Work ID	NNMV-2026-00125
Old Value	55%
New Value	68%
Timestamp	15-09-2026 10:30
Source	Web / Mobile / API

- Login and logout events
- Work creation/modification
- Progress updates
- Photo uploads
- Status changes
- User/role changes

20. Security Requirements

- Username/password authentication.
- Secure password hashing.
- Role-Based Access Control.
- HTTPS/TLS.
- Session timeout and secure session management.
- API authentication and authorization.
- File type/size validation.
- Secure photo/document storage.
- Audit logging.
- Regular database backup.
- Database access controls.
- Protection against common web vulnerabilities (OWASP-aligned practices).

Optional security enhancements: OTP/2FA, IP restrictions, device/session management and privileged-action approval.

21. Data Model & Database

Master Tables

- users
- roles
- permissions
- wards
- zones
- departments
- schemes
- financial_sources
- work_types
- contractors
- milestones_master

Transaction Tables

- development_works
- work_milestones
- progress_updates
- progress_photos
- work_documents
- work_assignments

System Tables

- notifications
- audit_logs
- login_logs
- system_settings

Core relationship: Ward → Development Works → Scheme/Financial Source/Department/JE/Contractor → Milestones → Progress Updates → Photos/Documents.

22. API & Portal Integration

यदि existing portals APIs उपलब्ध कराते हैं, तो direct integration preferred होगा। Suggested API contract:

Method	Endpoint Example	Purpose
GET	/api/works	Work list
GET	/api/works/{work_id}	Work details
GET	/api/wards	Ward master
GET	/api/schemes	Scheme master
GET	/api/departments	Department master
GET	/api/contractors	Contractor master
GET	/api/works/{work_id}/progress	Progress history
GET	/api/works/{work_id}/photos	Photo history
POST	/api/progress	Create progress update
POST	/api/progress/photos	Upload progress photo
PUT	/api/works/{id}/status	Update status
PUT	/api/milestones/{id}	Update milestone

Actual endpoint names existing API documentation और technical audit के बाद finalize किए जाएंगे।

23. Data Migration

दो existing portals के data को centralized model में consolidate करना project का critical workstream है।

36. Existing portal/database structure identify करना.
37. Available APIs और authentication mechanism identify करना.
38. Existing tables/fields और data types document करना.
39. Duplicate/overlapping works identify करना.
40. Wards और zones का standard mapping करना.
41. Schemes और financial sources normalize करना.
42. JE/Officer/Contractor mapping करना.
43. Existing Work IDs और historical progress identify करना.
44. Photos/documents का linkage verify करना.
45. Data migration scripts और validation reports तैयार करना.

Target architecture: Existing Portal 1 + Existing Portal 2 + Reference Data → Data Mapping/Standardization → Central Database → Monitoring Dashboard.

24. Non-Functional Requirements

Area	Requirement
Performance	Common dashboard/filter responses ideally within acceptable web application response time under expected load
Scalability	Architecture should support increasing wards, works, photos and users

Availability	Production hosting with backup and recovery strategy
Usability	Simple Hindi/English-friendly UI with clear status indicators
Responsive	Desktop-first officer dashboard + mobile-friendly JE screens
Maintainability	Modular backend, documented APIs, database migrations
Security	RBAC, secure authentication, validation, logging
Backup	Automated database and media backup
Observability	Application/error logs and operational monitoring

25. Development Phases

Phase	Deliverables
Phase 1 – Foundation	Database, authentication, RBAC, master management, work management
Phase 2 – Monitoring	JE dashboard, progress, milestones, photos, status
Phase 3 – Management	Central dashboard, ward/scheme/department analytics, reports
Phase 4 – Integration	nnmv.online, nnmvprojects.online, Construction Portal APIs/data migration
Phase 5 – GIS/Public	Map and public dashboard
Phase 6 – Advanced	Notifications, mobile/PWA enhancements, photo/GPS verification, advanced analytics

26. MVP Scope

Admin

- Login
- User management
- Ward/Scheme/Department masters
- Work management
- JE assignment

JE

- Login
- Assigned works
- Progress %
- Status
- Milestone
- Photos
- Remarks

Councillor

- Login
- Ward works

- Progress view
- Photos
- Status

Officer

- Central dashboard
- Filters
- Ward/Scheme/Department monitoring
- Delayed works

System

- Reports
- Audit log
- Secure photo storage

27. Acceptance Criteria

- हर work का unique Work ID हो.
- हर work किसी Ward से mapped हो.
- Scheme/Financial Source defined हो.
- JE केवल assigned works update कर सके.
- Progress update में date, percentage, status और remarks capture हों.
- Photo upload और category/milestone linkage काम करे.
- Milestone-wise progress उपलब्ध हो.
- Councillor अपने ward के works देख सके.
- Officers पूरे authorized Nagar Nigam data को monitor कर सकें.
- Advanced filtering उपलब्ध हो.
- Delayed works reliably identify हों.
- Important changes audit log में capture हों.
- Reports export हो सकें.
- Existing portals से agreed data successfully migrate/integrate हो.
- Role-based access का UAT में verification हो.

28. Inputs Required Before Development

A. Existing Data

- Current work list
- All wards
- Zones
- Departments
- JE list
- Councillor list
- Contractor list
- Schemes
- Financial sources
- Work types

B. Existing Portals

- nnmv.online database/API access
- nnmvprojects.online database/API access
- Construction Department portal access
- Existing login/role structure
- API documentation, if available

C. Work Data

- Work ID
- Work Name
- Ward
- Location
- Scheme
- Financial Source
- Work Type
- Sanction Amount
- Work Order
- Contractor
- JE
- Start Date
- Completion Date
- Current Status
- Current Progress
- Existing Photos

D. Business Rules

- Who can create a work?
- Who can approve progress?
- How often must JE update progress?
- When is a work considered Delayed?
- Who verifies Completion?
- Which fields are public?
- Can progress be edited after submission?
- How are revised completion dates handled?

29. Risks & Mitigations

Risk	Impact	Mitigation
Duplicate/dirty legacy data	Incorrect dashboard totals	Data profiling, normalization and validation before migration
No/limited APIs	Manual synchronization	Database integration or controlled import/export process
Inconsistent Ward/Scheme naming	Incorrect filtering/reporting	Master-data mapping and controlled vocabularies
Unverified progress	Misleading status	Photo evidence, audit trail and officer verification
Large photo volume	Storage/performance issues	Object storage, compression, thumbnails and retention policy
Role/access misconfiguration	Data exposure	RBAC matrix, least privilege and UAT
Late JE updates	Stale dashboard	Pending-update alerts and update compliance KPI

30. Recommended Implementation Approach

Development शुरू करने से पहले UI बनाने के बजाय Data & API Audit को first milestone रखना recommended है।

46. Existing Portal Audit
47. Database/API Documentation
48. Data Mapping & Standardization
49. Final Database Schema
50. UI/UX Wireframes
51. API Architecture
52. Backend + Frontend Development
53. Data Migration
54. UAT with Officers/JEs/Councillors
55. Production Deployment

Strategic Recommendation

Project का सबसे बड़ा implementation risk dashboard development नहीं बल्कि existing portals के data को एक standardized Central Development Works data model में सही तरीके से consolidate करना है। इसलिए data ownership, unique Work ID, Ward/Scheme master mapping और update/approval workflow को शुरुआत में freeze करना चाहिए।

Appendix A – Suggested Work Status Flow

Not Started → In Progress → Under Verification → Completed

Alternative exception states: In Progress → On Hold / Delayed / Cancelled

Appendix B – Example Road Work Progress

Stage	Example Progress State
Base Preparation	Completed
Gitti / Sand Work	Completed
Interlocking	In Progress
Finishing	Not Started
Completion	Pending

Appendix C – Suggested Dashboard KPI Set

KPI	Purpose
Total Works	Overall workload
Total Sanctioned Amount	Financial scale
Ongoing Works	Active execution
Completed Works	Completion volume
Delayed Works	Risk/escalation
Average Progress %	Overall execution health
Ward Completion %	Geographic performance
Scheme Completion %	Funding/program performance
JE Update Compliance	Field reporting discipline
Works Due for Completion	Near-term management focus